

Assume the following tables:

- students(id, name, department_id, marks)
- employees(emp_id, emp_name, salary, department_id)
- departments(department_id, department_name, location)
- orders(order_id, customer_id, amount)
- customers(customer_id, customer_name, city)

Single Row Subqueries

(Subquery returns only one value)

Question 1

Find students who scored more than the **average marks** of all students.

```
select id,name from students where marks > (select avg(marks) from students);
```

Question 2

Display employees whose salary is greater than the **minimum salary** in the company.

```
select emp_id,emp_name from employees where salary > (select min(salary) from employees);
```

Question 3

Find the student(s) who scored the **highest marks** in the class.

```
select id,name,marks from students where = (select max(marks) from students);
```

Multiple Row Subqueries

(Subquery returns multiple values)

Question 4

Find students who belong to departments located in '**Hyderabad**' (Assume departments table has location column).

select id,name from students where department_id in (select department_id from departments where location = 'Hyderabad');

Question 5

Display employees who work in departments named **'HR' or 'Sales'**.

select emp_id,emp_name from employees where department_id in (select department_id from departments where department_name in ('HR','sales'));

Question 6

Find customers who placed orders with an amount greater than 10,000.

select customer_id,customer_name from customers where customer_id in (select customer_id from orders where amount > 10000);

Multiple Column Subqueries

(Subquery returns more than one column)

Question 7

Find students who have the **same department and marks** as student with ID = 3.

select id,name from students where department_id,marks = (select department_id,marks from students where id = 3);

Question 8

Display employees who have the same **salary and department_id** as employee with emp_id = 5.

select emp_id,emp_name from employees where salary,department = (select salary,department_id from employees where emp_id = 5);

Nested Subqueries

(Subquery inside another subquery)

Question 9

Find employees who earn more than the **average salary of the department that has the highest number of employees**.

```
select emp_id,emp_name from employees salary > (select avg(salary) from
employees where department_id = (select department_id from employees group
by department_id order by count(*) desc limit 1));
```

Question 10

Find students who scored more than the **average marks of the department which has the highest average marks**.

```
select id,name from students where marks > (select avg(marks) from students
where department_id = (select department_id from students group by
department_id order by avg(marks) desc limit 1;
```

Correlated Subqueries

(Subquery depends on outer query)

Question 11

Find students who scored more than the **average marks of their own department**.

```
select id,name from students where marks > (select avg(marks) from students
where department_id = students.department_id);
```

Question 12

Display employees who earn more than the average salary in their department.

```
select emp_id,emp_name from employees salary > (select avg(salary) from
employees where department_id = employees.department_id);
```

Question 13

Find customers who placed **more than one order**.

```
select customer_id,customer_name from customers where customer_id in
(select customer_id from customers group by customer_id order by
count(order_id) > 1);
```